King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Final Exam Term 242

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The parametric curve $x = 4\sec t$, $y = 3\tan t$ is represented in rectangular form by the equation

(a) $\dfrac{x^2}{9} - \dfrac{y^2}{16} = 1$

(b) $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$

(c) $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$

(d) $y = x^2 - 144$

(e) $\dfrac{x}{4} + \dfrac{y}{3} = 1$

2. The maximum value of $f(x, y, z) = xyz$ on the sphere $x^2 + y^2 + z^2 = 12$ is equal to

(a) 10

(b) 16

(c) 8

(d) 12

(e) 27

3. The length of the polar curve $r = e^{3\theta}$, $0 \le \theta \le \dfrac{\pi}{3}$, is equal to

(a) $e^{\pi} - 1$

(b) $3\,(e^{\pi} - 1)$

(c) $\dfrac{\sqrt{10}\,e^{\pi}}{3}$

(d) $\dfrac{\sqrt{10}}{3}\,(e^{\pi} - 1)$

(e) $\sqrt{10}\,(e^{\pi} - 1)$

4. The volume of the solid bounded above by $z = 3 - x^2 - y^2$ and below by $z = 2x^2 + 2y^2$ is equal to

(a) $\dfrac{7\pi}{2}$

(b) 2π

(c) $\dfrac{5\pi}{2}$

(d) $\dfrac{3\pi}{2}$

(e) 3π

5. If $D = \{(x, y, z) : 0 \le z \le \sqrt{4 - x^2 - y^2},\ 0 \le y \le \sqrt{4 - x^2},\ -2 \le x \le 2\}$, then $\iiint_D \sin(x^2 + y^2 + z^2)\, dV =$

(a) $\int_0^{\pi} \int_0^{\pi} \int_0^2 \rho^2 \sin\rho \cdot \sin\phi \, d\rho \, d\phi \, d\theta$

(b) $\int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^2 \sin(\rho^2) \cdot \sin\phi \, d\rho \, d\phi \, d\theta$

(c) $\int_0^{2\pi} \int_0^{\pi} \int_0^4 \rho \sin(\rho^2) \cdot \sin\phi \, d\rho \, d\phi \, d\theta$

(d) $\int_0^{\pi} \int_0^{\pi/2} \int_0^4 \rho \sin(\rho^2) \cdot \sin\phi \, d\rho \, d\phi \, d\theta$

(e) $\int_0^{\pi} \int_0^{\pi/2} \int_0^2 \rho^2 \sin(\rho^2) \cdot \sin\phi \, d\rho \, d\phi \, d\theta$

6. Let $f(x, y) = 2x + y$ and $R = \{(x, y) : 0 \le x \le a,\ 0 \le y \le a\}$. If the average value of f over R is 6, then $a =$

(a) $\dfrac{12}{5}$

(b) 6

(c) $\dfrac{3}{2}$

(d) 4

(e) 2

7. The distance between the planes $\mathcal{P}_1 : 2x - y + 3z = 1$ and $\mathcal{P}_2 : -4x + 2y - 6z = 3$ is equal to

(a) $\dfrac{5}{2\sqrt{14}}$

(b) $\dfrac{1}{\sqrt{14}}$

(c) $\dfrac{2}{\sqrt{14}}$

(d) $\dfrac{1}{2\sqrt{14}}$

(e) $\dfrac{3}{\sqrt{14}}$

8. The graph of the function $f(x, y) = -2x^3 + 8xy - 4y^2 + 5$ has

(a) one saddle point and one relative maximum

(b) one relative minimum and one relative maximum

(c) one saddle point and one relative minimum

(d) one saddle point and two relative maxima

(e) no saddle point

9. If $w = x^2 + y^2 + z^2$, $x = t\sin(2r)$, $y = t\cos(2r)$, $z = rt^2$, then the value of $\dfrac{\partial w}{\partial r}$ when $r = \dfrac{\pi}{8}$ and $t = 1$ is equal to

(a) $\dfrac{\pi}{8}$

(b) 0

(c) $\dfrac{\pi}{4}$

(d) $2 + \dfrac{\pi}{4}$

(e) $2 + \dfrac{\pi}{8}$

10. If $R = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le \ln 3\}$, then $\iint_R ye^{xy}\, dA =$

(a) $2 - \ln 3$

(b) $4 - \ln 3$

(c) $1 + \ln 3$

(d) $e - \ln 3$

(e) $3 - \ln 3$

11. If (a, b, c), with $a < 0$, is the point on the surface $x^2 + 4y^2 - z^2 = -4$ where the tangent plane is parallel to the plane $2x - 4y + 3z = 1$, then $a + b + c =$

(a) 6

(b) 4

(c) -1

(d) 0

(e) -3

12. Find $\lim_{(x,y)\to(0,0)} \dfrac{y}{x^2+y^2}$, if it exists.

(a) 0

(b) DNE

(c) 2

(d) -1

(e) $\dfrac{1}{2}$

13. Which one of the following points lies on the line that passes through the point $(2,3,4)$ and is perpendicular to the plane given by $3x+2y-z=6$?

(a) $(-1,1,5)$

(b) $(4,5,3)$

(c) $(0,0,0)$

(d) $(16,8,0)$

(e) $(8,7,1)$

14. If $z = \arctan\left(\dfrac{y}{x}\right)$, then $z_{xx} - z_x z_y + z_{yy} =$

(a) $\dfrac{x-y}{(x^2+y^2)^2}$

(b) $\dfrac{xy}{(x^2+y^2)^2}$

(c) $\dfrac{5xy}{(x^2+y^2)^2}$

(d) 0

(e) $\dfrac{3xy}{(x^2+y^2)^2}$

15. The volume of the solid lying in the first octant and bounded by the graphs of $z = 4-y^2$, $y = x$, $y = 1$ is equal to

(a) $\dfrac{11}{12}$

(b) 2

(c) $\dfrac{7}{4}$

(d) $\dfrac{5}{12}$

(e) $\dfrac{9}{14}$

16. $\int_0^2 \int_{x^2}^{2x} f(x,y)\,dy\,dx =$

(a) $\int_0^2 \int_y^{2\sqrt{y}} f(x,y)\,dx\,dy$

(b) $\int_0^4 \int_{y/2}^{\sqrt{y}} f(x,y)\,dx\,dy$

(c) $\int_0^4 \int_y^{2y^2} f(x,y)\,dx\,dy$

(d) $\int_0^2 \int_{y/2}^{\sqrt{y}} f(x,y)\,dx\,dy$

(e) $\int_{x^2}^{2x} \int_0^2 f(x,y)\,dx\,dy$

17. The maximum value of $f(x,y) = 3x^2 + 2y^2 - 4y$ over the region in the xy-plane bounded by the graphs of $y = x^2$ and $y = 4$ is equal to

(a) 0

(b) 28

(c) 16

(d) -2

(e) 20

18. If $R = \{(x,y) : x^2 + y^2 \leq 4,\ x \geq 0\}$, then $\iint_R e^{-x^2-y^2}\,dA =$

(a) $\frac{\pi}{4}\left(1 - e^{-4}\right)$

(b) $\frac{\pi}{2}\left(2 + e^{-4}\right)$

(c) $\frac{\pi}{4}\left(1 + e^{-4}\right)$

(d) $\pi\left(1 - e^{-4}\right)$

(e) $\frac{\pi}{2}\left(1 - e^{-4}\right)$

19. Let $f(x,y) = 3x^2 - xy^2 + y^3$, $P(-1,1)$ and $Q(2,-2)$. The directional derivative of f at P in the direction of $\overrightarrow{PQ}$ is equal to

(a) $4\sqrt{2}$

(b) $2\sqrt{2}$

(c) $-3\sqrt{2}$

(d) $-6\sqrt{2}$

(e) $-\frac{\sqrt{2}}{3}$

20. If (a, b, c), with $c > 0$, is the point on the cone $z^2 = x^2 + y^2$ that is closest to the point $(1, 2, 0)$, then $8abc =$

(a) $-3\sqrt{5}$

(b) $4\sqrt{5}$

(c) $2\sqrt{5}$

(d) $\sqrt{5}$

(e) $5\sqrt{5}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\sec^2 t - \tan^2 t = 1$ gives $\frac{x^2}{16} - \frac{y^2}{9} = 1$.
2	**(c)**	Lagrange gives $x^2 = y^2 = z^2 = 4$, so $\max xyz = 2 \cdot 2 \cdot 2 = 8$.
3	**(d)**	$L = \int_0^{\pi/3} \sqrt{10} e^{3\theta} d\theta = \frac{\sqrt{10}}{3}(e^{\pi} - 1)$.
4	**(d)**	Surfaces meet at $r = 1$: $V = \int_0^{2\pi} \int_0^1 3(1 - r^2) r \, dr \, d\theta = \frac{3\pi}{2}$.
5	**(e)**	Quarter ball: $0 \le \rho \le 2$, $0 \le \phi \le \frac{\pi}{2}$, $0 \le \theta \le \pi$; $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$.
6	**(d)**	Average $= \frac{1}{a^2} \cdot \frac{3a^3}{2} = \frac{3a}{2} = 6$, so $a = 4$.
7	**(a)**	Point $(0, -1, 0)$ on $\mathcal{P}_1$: $\frac{\lvert -2-3 \rvert}{\sqrt{56}} = \frac{5}{2\sqrt{14}}$.
8	**(a)**	Critical points: $(0, 0)$ saddle, $(\frac{4}{3}, \frac{4}{3})$ relative max.
9	**(c)**	$w = t^2 + r^2 t^4$, so $w_r = 2rt^4 = \frac{\pi}{4}$.
10	**(a)**	Inner $\int_0^1 y e^{xy} dx = e^y - 1$; then $[e^y - y]_0^{\ln 3} = 2 - \ln 3$.
11	**(b)**	$\nabla F \parallel \langle 2, -4, 3 \rangle$: point $(-4, 2, 6)$, sum 4.
12	**(b)**	Path $y = 0$ gives 0, path $x = 0$ gives $1/y$ (unbounded): DNE.
13	**(a)**	Line $(2 + 3t, 3 + 2t, 4 - t)$; $t = -1$ gives $(-1, 1, 5)$.
14	**(b)**	$z_{xx} + z_{yy} = 0$ and $-z_x z_y = \frac{xy}{(x^2+y^2)^2}$.
15	**(c)**	$V = \int_0^1 \int_0^y (4 - y^2) dx \, dy = 2 - \frac{1}{4} = \frac{7}{4}$.
16	**(b)**	Region between $y = x^2$ and $y = 2x$: $0 \le y \le 4$, $\frac{y}{2} \le x \le \sqrt{y}$.
17	**(b)**	Interior value -2; on $y = 4$, $f = 3x^2 + 16$ peaks at $(\pm 2, 4)$: maximum 28.
18	**(e)**	Polar: $\int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} r \, dr \, d\theta = \frac{\pi}{2}(1 - e^{-4})$.
19	**(d)**	$\nabla f(P) = \langle -7, 5 \rangle$, $\mathbf{u} = \frac{1}{\sqrt{2}}\langle 1, -1 \rangle$: $-6\sqrt{2}$.
20	**(c)**	Minimize $(x-1)^2 + (y-2)^2 + x^2 + y^2$: $(\frac{1}{2}, 1, \frac{\sqrt{5}}{2})$, so $8abc = 2\sqrt{5}$.